

How Human Experts Educate Specialized LLMs: Filling Knowledge Gaps in KG-Augmented Generation through Hallucination Detection

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Abstract

The integration of Domain Knowledge Graphs (DKG) into Retrieval-Augmented Generation has emerged as a promising approach for constructing Specialized Large Language Models (spLLMs). On account of the scarcity of high-quality DKGs, existing approaches employ an evolutionary framework, wherein the DKG is continuously evolved alongside its utilization for enhancing the LLM. Yet, these methods face two key limitations: 1) heavy reliance on costly expert knowledge, and 2) neglect of the connection between the LLM's inherent knowledge and external expert knowledge. To address these issues, this paper introduces Epistemic Cognition-enhanced Specialized LLMs (EC-spLLM), a novel evolutionary framework that instills epistemic cognition into the LLM to systematically exploit both its internal knowledge and expert knowledge. At the core of EC-spLLM lies the Hallucination Detection-based Epistemic Cognition (HDEC) mechanism, which assesses the reliability of LLM-generated responses using the LLM's self cognition and hallucination detection. This assessment ability enables EC-spLLM to selectively adopt either the expert-provided golden answer or a reliable LLM-generated answer during DKG evolution, thereby reducing dependence on experts and bridging internal and external knowledge sources to enhance performance. We conducted extensive experiments on five datasets spanning five domains, e.g.,

emotional sociology, biology, etc. Results show that EC-spLLM reduces the usage of golden answers by an average of 67% while retaining 97.2% of the accuracy achieved by the SOTA method, and outperforms all other baselines.

CCS Concepts

• Computing methodologies → Artificial intelligence; • Theory of computation → Semantics and reasoning.

Keywords

Specialized large language models, domain knowledge graph, retrieval-augmented generation, hallucination detection, social network

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1 Introduction

Since the debut of GPT-3.5, Large Language Models (LLMs) have exhibited remarkable capabilities across diverse tasks, including text generation, comprehension, and reasoning [41]. However, LLMs often struggle with reasoning tasks demanding specialized expertise [23]. To address this limitation, Retrieval-Augmented Generation (RAG) has been explored to augment LLMs with external domain knowledge, thereby enhancing their performance on specialized tasks [16, 21, 22]. RAG-based specialized LLMs operate through a *query-retrieve-generate* framework: Upon receiving a specialized query, the system retrieves relevant information from an external

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domain-specific knowledge source, and then incorporates it into input prompts to guide the LLM for response generation. Notably, RAG constructs specialized LLMs without the requirement of model retraining [30] or parameter fine-tuning [9], which entail substantial efforts for the acquisition of large-scale instruction data and complex design of training pipeline.

Despite these benefits, RAG carries the risk of introducing irrelevant or misleading information [35], leading to the issue of knowledge noise. To mitigate this problem, state-of-the-art RAG approaches leverage knowledge graphs (KGs) as external knowledge sources. KGs provides structured, explicit, and high-quality knowledge that enhances reasoning capabilities and captures complex dependencies often missed in unstructured textual data. The performance of KG-augmented LLM is highly dependent on the quality and comprehensiveness of the underlying KGs. Existing KG-augmented LLMs [18, 26] mostly rely on generalized KGs, e.g., Wikidata [28] and Freebase [1]. However, these generalized KGs contain coarse-grained and domain-agnostic knowledge, which limits their effectiveness in addressing specialized queries. Using Domain Knowledge Graphs (DKGs) is a straightforward solution to the above challenge. Unfortunately, high-quality DKGs are only available in a limited number of fields, such as medicine [3] and commerce [15]. To address this issue, Way-to-Specialist (WTS [39]) proposes an evolutionary framework for KG-augmented LLMs. It continuously evolves incomplete DKGs while implementing KG-augmented LLMs for solving specialized tasks. This capability is also particularly vital for dynamic environments like social networks, where user interests and trending topics evolve rapidly. In such contexts, WTS can capture real-time changes in user sentiment and emerging narratives, updating the underlying knowledge structure to reflect the latest states.

Nevertheless, the current evolutionary framework suffers from two fundamental limitations. **Firstly**, it relies heavily on golden answers for specialized tasks to extract new knowledge triples and evolve the DKG. Acquisition of these golden answers typically requires expert analysis, which is both time-consuming and labor-intensive. **Secondly**, the current framework relies solely on expert knowledge to evolve the DKG, resulting in a disconnect between expert-derived knowledge and the intrinsic knowledge of LLM. This issue impedes the effective answering of specialized queries.

To address the aforementioned challenge, we introduce an epistemic cognition mechanism for KG-augmented specialized LLMs. This mechanism enables them to assess their knowledge boundaries and determine whether they can provide a reliable answer or should seek assistance from human experts. When the specialized LLM recognizes that a question lies outside its knowledge boundaries, it can solicit golden answers from human experts and extract knowledge from these answers to fill the knowledge gaps in the DKG. Conversely, when the model exhibits high confidence in its generated response, it can extract knowledge from that response to complete the DKG, thus contributing to bridging the gap between expert knowledge and the intrinsic knowledge of the LLM. In this way, the specialized LLM requests expert assistance only when it recognizes its own incapability, rather than doing so indiscriminately every time, which significantly reduces its reliance on human experts.

However, LLMs exhibit a limited capacity of self cognition, and they may provide incorrect information in response without

self-doubt to questions they cannot reliably answer [36, 37]. To address this issue, we draw inspiration from hallucination detection techniques [4–7, 19, 38, 42] and design the Hallucination Detection-based Epistemic Cognition (HDEC) mechanism. In HDEC, responses for which the LLM expresses self-doubt are immediately flagged as unreliable, whereas confident responses undergo a subsequent hallucination detection process. This dual approach enables HDEC to identify instances where an LLM confidently asserts false information, providing more robust reliability assessments.

Centered around the HDEC mechanism, we propose a novel evolutionary framework for KG-augmented LLMs called Epistemic Cognition-enhanced Specialized LLMs (**EC-spLLM**). By leveraging HDEC, EC-spLLM gains the capability to assess the reliability of its own responses. On the one hand, this capability allows it to retrieve external knowledge in a compound interest manner when self-doubt or hallucination is detected, thereby improving response accuracy. On the other hand, it empowers the system to make informed decisions on whether to rely on its own outputs or defer to golden answers during the dynamic evolution of the DKG. As a result, the model's dependence on human experts is substantially reduced, offering a cost-efficient solution that enhances both the autonomy and accuracy of KG-augmented specialized LLM systems. The contributions of this paper are summarized as follows:

1. We propose a novel evolutionary framework for KG-augmented LLMs called EC-spLLM, whose core is the HDEC mechanism. Guided by HDEC's assessments, EC-spLLM selectively avoids relying on golden answers when its own answers are deemed reliable, thereby significantly reducing reliance on experts. Furthermore, driven by HDEC's outputs, EC-spLLM is able to autonomously perform compound interest retrieval, which further enhances its accuracy.
2. HDEC innovatively employs hallucination detection techniques to enhance the epistemic cognition of LLMs. Drawing inspiration from semantic entropy—a representative hallucination detection approach—it analyzes the probability distribution of LLM's outputs in semantic space to identify patterns indicative of self-doubt or hallucinations. HDEC can automatically assess the reliability of LLM's output without prior knowledge and can be seamlessly integrated into KG-augmented specialized LLM systems to improve accuracy and autonomy.
3. We conduct extensive experiments on five datasets across five domains—including four specialized fields and one general domain—to assess the performance of EC-spLLM. The results demonstrate that on average, EC-spLLM reduces the usage of golden answers by approximately 67% while maintaining 97.2% of the SOTA accuracy and outperforming all other baselines, thus validating its effectiveness and versatility.

2 Related Work

RAG-based Specialized LLM. RAG-based specialized LLMs aim to leverage knowledge from an external domain-specific knowledge source to enhance an LLM's reasoning capabilities in a particular domain [16, 21, 22]. Depending on the structural form of the data source, such systems can be broadly categorized into Text-based Specialized LLMs [16, 29] and KG-based Specialized LLMs [21, 33]. Because KGs provide more structured knowledge and

clearer reasoning paths, KG-based Specialized LLMs have increasingly attracted researchers' attention. However, their performance is constrained by the quality of the DKG they rely on, and high-quality DKGs are available for only a limited number of domains (e.g., medicine [3] and commerce [15]). To overcome this limitation, some studies [17, 22, 39] have proposed evolutionary frameworks for KG-augmented Specialized LLMs that simultaneously enhance LLMs using existing DKGs while refining and expanding the DKGs themselves. While promising, these frameworks are constrained by their heavy dependence on expert-provided golden answers, which are costly to acquire in terms of time and labor. Furthermore, when evolving the DKG, they exclusively leverage this external expert knowledge, thereby neglecting the rich intrinsic knowledge available within the LLM itself.

Hallucination Detection. In the context of LLMs, the term hallucination typically refers to the generation of content that is non-sensical or inconsistent with the given source material [8, 10, 20]. Current hallucination detection methods generally fall into three main categories: 1) *Internal-State-based Detection*: These methods aim to train a model using a combination of the LLM's internal state data (e.g., hidden layer activations) and the ground-truth correctness of the corresponding answers [4, 5, 13]. The model is expected to predict whether the LLM will hallucinate based on its internal states. However, these methods are not applicable to black-box LLMs and often lack interpretability. 2) *Retrieval-based Detection*: These approaches focus on comparing the responses generated by the LLM with retrieved factual knowledge to determine whether hallucinations occurred [7, 25, 38]. While straightforward and easy to understand, retrieval-based methods rely heavily on prior knowledge, making them impractical in scenarios where answers cannot be pre-determined. 3) *Sampling-based Detection*: The core idea behind these methods is to generate multiple answers from the LLM for the same context, then quantify the inconsistency among these answers to identify hallucinations [6, 19, 24]. Each approach introduces its own metric to evaluate the level of inconsistency in the generated responses. Among these, the method most relevant to this work is Semantic Entropy (SE). SE clusters LLM-generated answers in semantic space, computes their probabilities, and calculates an entropy value to quantify semantic uncertainty. SE has shown strong performance across multiple datasets and is effective for hallucination detection. However, when evaluating whether the LLM can reliably answer a specific query, SE does not account for the self-awareness of the LLM.

3 Methodology

3.1 Education Pipeline of Specialized LLMs

The overall pipeline of our method comprises three phases: **DKG-augmented Generation**, **Hallucination Detection-based Epistemic Cognition (HDEC)**, and **LLM-assisted DKG Evolution**.

1) As illustrated in Figure 1, when a query is posed, it proceeds first to the DKG-augmented generation phase. In this phase, key entities are extracted from the question and utilized to retrieve relevant knowledge from the current DKG. The retrieved knowledge is subsequently pruned and incorporated into tailored prompts to support answer reasoning.

2) Simultaneously with the reasoning process, the HDEC phase is activated. It begins by sampling multiple candidate responses and clustering them based on semantic similarity. HDEC then identifies responses expressing self-doubt and leverages their cumulative probability to evaluate whether the LLM is competent to address the query. If this assessment is positive, HDEC further detects potential hallucinations in the responses. A negative result in either cognitive check above (self-doubt or hallucination) causes HDEC to return a negative assessment and trigger a fallback to Phase 1 for further knowledge retrieval. If both checks are passed, HDEC issues a positive assessment, permitting the pipeline to advance to the Phase 3. However, if the maximum retrieval depth has been reached, EC-spLLM advances to the next phase regardless of any assessment by HDEC.

3) The final phase is LLM-assisted DKG evolution. In this phase, if HDEC's ultimate evaluation is positive, EC-spLLM adopts the LLM's response as the system's output. Otherwise, EC-spLLM defers to expert assistance, using the golden answer provided by the expert as the final output. Irrespective of the output source, the knowledge contained within is extracted by this phase and integrated into the DKG after redundancy filtering, thereby evolving the DKG over time.

3.2 Phase1: DKG-augmented Generation.

The DKG-Augmented Generation consists of four stages: entity extraction, knowledge retrieval, knowledge pruning, and knowledge-augmented reasoning.

Entities Extraction: This stage leverages the LLM to extract entities from the question q . Specifically, q is loaded into a designed entity extraction prompt and passed to the LLM, which then produces a list of entities: $\mathcal{E}_q = [e_{q,1}, e_{q,2}, \dots, e_{q,i}]$.

Knowledge Retrieval: The knowledge retrieval stage employs a multi-layer iterative strategy. Here, we illustrate the process of layer- d retrieval ($d \geq 1$) as an example for comprehensible exposition and let $\mathcal{E}_q^{[d]}$ denote the input entity set for layer d . For layer $d = 1$, the input entity set $\mathcal{E}_q^{[1]} = \mathcal{E}_q$.

To acquire knowledge relevant to $\mathcal{E}_q^{[d]}$, EC-spLLM needs to interact with a DKG. A DKG is a structured knowledge base designed to represent entities and the intricate relationships between them. Formally, we define the DKG $\mathcal{G}$ as a set of factual triplets: $\mathcal{G} = \{(e_s, r, e_o) \mid e_s, e_o \in \mathcal{E}, r \in \mathcal{R}\}$. Here, each triplet (e_s, r, e_o) represents a relationship where e_s and e_o denote the subject and object entities, respectively, and r represents the relation linking them. $\mathcal{E}$ is the set of all entities, and $\mathcal{R}$ is the set of all relations in the graph. In the EC-spLLM framework, to enable efficient semantic retrieval, the DKG is stored in a vector database. All entities and relations are mapped to high-dimensional vectors through an embedding function $\mathcal{F}(\cdot)$, allowing their semantic similarity can be measured via vector distance calculations.

For each entity $e \in \mathcal{E}_q^{[d]}$, retrieval proceeds in two steps:

1) *Semantic Matching*. This step retrieves e in $\mathcal{G}$ and collects triples including e . To address synonymy and terminological variation in evolving domains (e.g., rapidly changing LLM terminology), we perform fuzzy search instead of exact matching. Experimental results presented in Appendix E demonstrate the effectiveness of

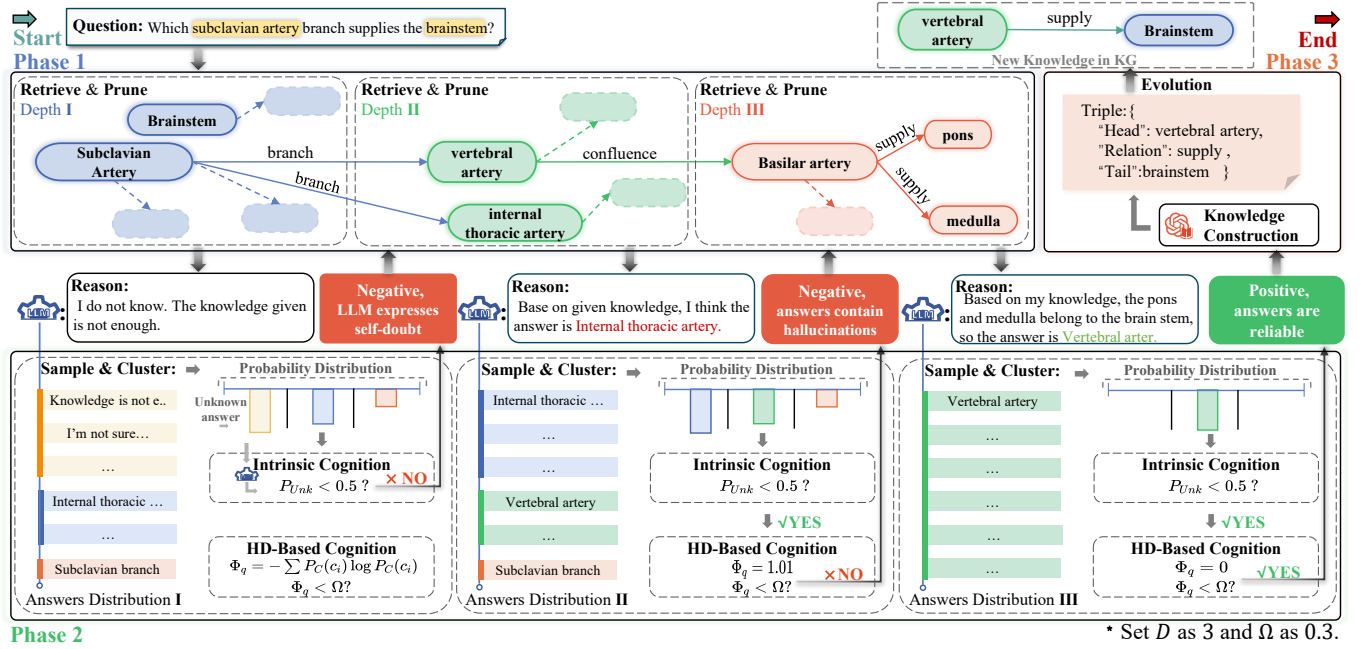


Figure 1: Overview of EC-spLLM.The top-left, bottom, and top-right sections denote Phase 1, Phase 2, and Phase 3, respectively, corresponding to DKG-augmented Generation, Hallucination Detection-based Epistemic Cognition, and LLM-assisted DKG Evolution in sequence.

this strategy. The semantic similarity between entity e and candidate entities e_k is computed via cosine distance in the embedding space:

$$\text{Dis}(e, e_k) = 1 - \frac{\mathcal{F}(e)^\top \mathcal{F}(e_k)}{\|\mathcal{F}(e)\| \cdot \|\mathcal{F}(e_k)\|}. \quad (1)$$

For each entity $e \in \mathcal{E}_q^{[d]}$, define the candidate knowledge set as $\mathcal{K}_e^{\text{cand}} = \{k \in \mathcal{G} \mid \mathcal{M}(e, k)\}$ and define $\mathcal{K}_q^{[d], \text{SM}} = \bigcup_{e \in \mathcal{E}_q^{[d]}} \mathcal{K}_e^{\text{cand}}$ as the output of this step, where $\mathcal{M}(e, k)$ denotes the semantic matching condition $\text{Dis}(e, e_s) \geq \theta \vee \text{Dis}(e, e_o) \geq \theta$, with $e_s, e_o \in k$ and θ represents a predefined similarity threshold.

2) *Similarity Filtering*. To prioritize knowledge relevant to the original question q , we compute question-knowledge relevance using cosine similarity as $\text{Sim}(q, k) = \frac{\mathcal{F}(q) \cdot \mathcal{F}(k)}{\|\mathcal{F}(q)\| \cdot \|\mathcal{F}(k)\|}$, $\forall k \in \mathcal{K}_q^{[d], \text{SM}}$. The knowledge with $\text{Sim}(q, k)$ higher than the threshold λ is retained. Then the output of this step is defined as $\mathcal{K}_q^{[d], \text{SF}} = \{k \in \mathcal{K}_q^{[d], \text{SM}} \mid \text{Sim}(q, k) \geq \lambda\}$, which is also the final result $\mathcal{K}_q^{[d]}$ of the knowledge retrieval stage in depth d .

Depending on the assessment produced by Phase 2 (HDEC), the system may re-enter this stage to perform additional retrieval layers, until the preset depth threshold D is reached. If the retrieval proceeds to the next layer $d+1$, $\mathcal{E}_q^{(d+1)}$ will be set as $\{e \in \mathcal{K}_q^{[d]} \mid e \notin \bigcup_{i=1}^d \mathcal{E}_q^{(i)}\}$.

Knowledge Prune: As the knowledge graph evolves and deeper retrieval layers are explored, the retrieved knowledge tends to become increasingly verbose and noisy. To address this issue, we introduce a pruning mechanism based on LLM-guided scoring. Specifically, the retrieved knowledge set $\mathcal{K}_q^{[d]}$ in layer d is passed

to the LLM via designed prompt, where the LLM is asked to evaluate the usefulness of each item $k \in \mathcal{K}_q^{[d]}$ with respect to answering the original question q . The LLM assigns a score $g \in [0, 1]$ to each knowledge triple, where a score of 1 indicates that the information alone suffices to answer q , and a score of 0 indicates that it is entirely useless.

Finally, EC-spLLM selects the top- w knowledge triples with the highest scores as the output $\tilde{\mathcal{K}}_q^{[d]}$ of this pruning stage. In addition, based on the filtered results from the knowledge prune stage, the set of input entities for the next retrieval layer is updated as $\mathcal{E}_q^{[d+1]} = \{e \in \mathcal{K}_q^{[d]} \mid e \notin \bigcup_{i=1}^d \mathcal{E}_q^{(i)}\}$. This update ensures that retrieval resources are not wasted on entities previously explored or deemed unhelpful for answering the question.

Knowledge-augmented Reasoning: In this stage, the knowledge pruned in the previous stage is combined with the question q and fed to the LLM for final reasoning. Notably, to ensure the LLM can fully capture the relations among knowledge triples, all knowledge pruned from every retrieval layer up to the current depth is aggregated and presented together. Thus, the knowledge input provided to the LLM in layer d is $\tilde{\mathcal{K}}_q^{[d]} = \bigcup_{i=1}^d \tilde{\mathcal{K}}_q^{(i)}$. Using the combined knowledge, the LLM performs reasoning at the current layer and generates an output denoted as $a_q^{[d]}$.

3.3 Phase2: Hallucination Detection-based Epistemic Cognition.

The HDEC phase unfolds through four key stages: answer sampling, answer clustering, intrinsic cognition, and hallucination detection-based cognition.

Answer Sampling : During this stage, the LLM is queried u times using the same context employed in the knowledge-augmented reasoning stage, with a temperature parameter set to 1. This produces a sequence of answers $\mathcal{S} = \{s_1, s_2, \dots, s_u\}$, while HDEC simultaneously records the probability of each token in every generated answer.

As is well known, LLMs generate responses token by token [2]. Given a specific context x , the probability of a complete answer $s \in \mathcal{S}$ can be computed as the product of its token probabilities. The logarithmic form of this probability is expressed as:

$$\log P(s|x) = \sum_{i=1}^n \log P(t_i|t_{<i}, x), \quad (2)$$

where n is the number of tokens in s , t_i represents the i -th token in s , and $t_{<i}$ denotes the set of previous tokens.

However, this probability formulation unfairly penalizes long answers, as longer sequences inherently exhibit lower probabilities due to the conditional independence of tokens. To address this, HDEC applies *length normalization*, adjusting the probability of each answer as $\frac{1}{n} \sum_{i=1}^n \log P(t_i|t_{<i}, x)$.

Finally, HDEC combines each answer in $\mathcal{S}$ with its normalized probability to output a set of answer-probability pairs $\langle \mathcal{S}, P \rangle = \{ \langle s, p \rangle \mid s \in \mathcal{S}, p = \frac{1}{n} \sum_{i=1}^n \log P(t_i|t_{<i}, x) \}$, which forms the output of this stage.

Answer Clustering : To enable effective clustering, we adopt the notion of *entail* [6]. Specifically, given two sentences A and B , we define that A entails B if the truth of A logically implies the truth of B . When two answers mutually entail—for example, "Paris is the capital of France" and "The capital of France is Paris"—they are considered semantically equivalent and thus should therefore be assigned to the same cluster. Formally, for two answers $s_i, s_j \in \mathcal{S}$, we define their semantic equivalence relation based on mutual entailment as:

$$s_i \equiv s_j \iff \text{Entail}(s_i, s_j) \wedge \text{Entail}(s_j, s_i), \quad (3)$$

where $\text{Entail}(a, b)$ is a binary predicate indicating whether sentence a entails sentence b .

Using the well-designed prompt, the LLM is employed to identify such equivalence relations in $\mathcal{S}$, resulting in a cluster set $\mathcal{C} = \{C_1, C_2, \dots\}$. The probability of each cluster is then computed as $P(C_i) = \frac{\sum_{s_j \in C_i} P(s_j)}{\sum_{s \in \mathcal{S}} P(s)}$. Then the cluster probabilities, together with the clusters themselves, constitute the output $\langle \mathcal{C}, P_C \rangle$ of this stage.

Intrinsic Cognition: Previous studies [37] indicate that LLMs are capable of recognizing the boundaries of their knowledge, and may express self-doubt in response to questions beyond their scope. To leverage this, HDEC uses LLM to identify clusters in $\mathcal{C}$ that convey epistemic self-doubt—e.g., statements such as "I don't know" or "There is insufficient information". We refer to these as *unknown answers* C_{Unk} , and denote their total probability as $P_{\text{Unk}} = \sum_{C_i \in C_{\text{Unk}}} P(C_i)$.

Similar to the answers distribution I in Figure 1, if $P_{\text{Unk}} > 0.5$, the LLM is inferred to be more inclined to believe it cannot answer the question based on its current knowledge. This will lead to a negative assessment from the HDEC phase, prompting the whole system to return to the knowledge retrieval stage for deeper retrieval. Conversely, if $P_{\text{Unk}} \leq 0.5$, HDEC will proceed to the next stage.

Hallucination Detection-based Cognition: The self-awareness capabilities of LLMs are not entirely reliable [37], as hallucinations can sometimes cause the model to generate wrong responses without self-doubt. To mitigate the impact of hallucinations on HDEC's performance, we introduce the hallucination detection-based cognition stage. Motivated by the advantages of Semantic Entropy (SE [6]) in applicability and interpretability, this stage employs an SE-centered mechanism to realize hallucination detection.

SE is a method developed to quantify the level of confusion in an LLM's outputs, enabling the detection of hallucinations within the model's responses. It operates on the principle of measuring response inconsistency by computing the entropy of generated outputs within a semantic space. The calculation of the semantic entropy Φ_q of a question q typically involves the following steps: 1) Sampling multiple answers from the LLM for the question q , resulting in a candidate answer set $\{s_1, s_2, \dots, s_M\}$; 2) Clustering these answers into different semantic clusters $\{C_1, C_2, \dots, C_K\}$ based on semantic equivalence; 3) Calculating the probability of each semantic cluster C_i as the sum of the probabilities of all answers within the cluster $P(C_i|q) = \sum_{s_j \in C_i} P(s_j|q)$ and finishing the normalization; 4) Finally, calculating the semantic entropy using the Shannon entropy formula:

$$\Phi_q = H(C) = - \sum_{i=1}^K P(C_i|q) \log P(C_i|q). \quad (4)$$

A high value of Φ_q indicates that the model is hallucinating, whereas a low value of Φ_q suggests that the model's responses are consistent and reliable.

In this stage, HDEC uses the cluster-probability pairs $\langle \mathcal{C}, P_C \rangle$ from the previous stage to calculate the semantic entropy Φ_q of the LLM's responses to question q , as defined by Equation (4). If Φ_q exceeds a predefined threshold Ω , HDEC concludes the LLM has hallucinated. For instance, answer distribution 2 in Figure 1 exhibits a high Φ_q . This triggers a negative assessment from the HDEC phase, causing EC-spLLM to revert to the knowledge retrieval stage for additional knowledge to mitigate the hallucination. Conversely, if Φ_q is less than or equal to the threshold Ω (as exemplified by answers distribution 3 in Figure 1), HDEC deems the LLM's responses to be trustworthy. HDEC subsequently yields a positive assessment, and EC-spLLM enters the next phase.

3.4 Phase3: LLM-assisted DKG Evolution.

This phase consists of two key stages: knowledge construction and DKG evolution.

Knowledge Construction: This stage is responsible for extracting knowledge from the question q and its corresponding answer α to generate structured knowledge triples $\mathcal{K}_q^+$ which are used to evolve the DKG. The content of α is determined by the outcome of the HDEC phase:

1) *Negative Assessment:* As depicted in Figure 2, if the previous phase yields a negative assessment at the maximum depth D (due to self-doubt or hallucination), this stage will solicit a golden answer α_G from human experts. This expert-provided answer then becomes the knowledge source (i.e., $\alpha = \alpha_G$).

2) *Positive Assessment:* A positive assessment from HDEC at any depth allows this stage to use the LLM's own answer α_{LLM} as the

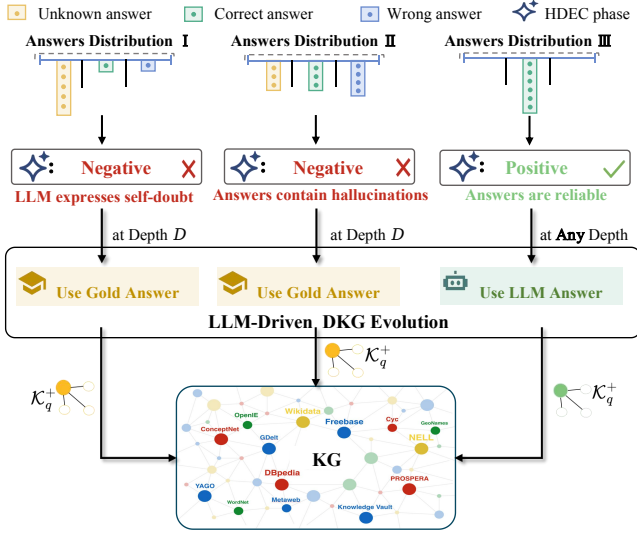


Figure 2: Selection between the golden answer and the LLM answer by LLM-Driven DKG Evolution under different assessments of the HDEC phase.

knowledge source (i.e., $\alpha = \alpha_{LLM}$). This bridges the gap between curated expert knowledge and the LLM’s intrinsic knowledge.

The resulting question-answer pair $\langle q, \alpha \rangle$ is then formatted into prompt and provided to the LLM for the generation of structured knowledge triples $\mathcal{K}_q^+$.

DKG Evolution: The knowledge $\mathcal{K}_q^+$ constructed during the last stage is further considered for integration into the DKG during this stage. However, it is worth noting that $\mathcal{K}_q^+$ may already exist within the current DKG, and redundant insertion could lead to unnecessary duplication. To address this issue, DKG evolution performs a retrieval process over the existing DKG to check for the presence of the knowledge triples in $\mathcal{K}_q^+$. If an identical knowledge triple is already present, this stage discards it; otherwise, it incorporates the new knowledge into the DKG.

4 Experiments and Discussions

4.1 Experiment Setup

4.1.1 Datasets and Models. We evaluate all methods on five Q&A datasets spanning diverse domains: Biology (BioASQ [27]), Emotional Sociology (TweetQA [34]), Science (SimpleQA [31]), Medicine (PubMedQA [11]), and a general domain dataset (NQ-open [14]). To better simulate real-world expert QA scenarios while controlling for knowledge redundancy, we rephrased the QA data and mixed them with the original samples. Details about the datasets and rephrasing procedure can be found in Appendix B. Golden answer usage (GA usage) and accuracy are used as the primary metrics to assess performance across datasets. Notably, for open-domain question answering datasets like NQ-open, where semantically equivalent answers may vary in surface form, we utilize GPT-4.1 as an automatic evaluator to assess the consistency between model outputs and the reference answers. For each method, we conducted experiments using both GPT-4.1 (2025-04-14) and DeepSeek-v3 (2025-03-24) on

all datasets. The two models were accessed respectively via the OpenAI API and the DeepSeek API.

4.1.2 Baselines and Hyper-parameters. We compare EC-spLLM against four prompting-based SOTA baselines: Zero-shot-CoT [12], CoT [32], ToG [26], and WTS [39], as well as a direct LLM querying baseline (hereafter referred to as Query Directly). For all methods, the inference temperature is set to 0.3 when they are asked to output the final result, and responses are required to be in JSON format to facilitate structured analysis. For ToG, we use Freebase [1] as the static knowledge graph, while both WTS and EC-spLLM are initialized with empty vector databases as their DKGs, implemented using ChromaDB¹ along with Qwen3-Embedding-0.6B [40] as the embedding model. For hyperparameter configurations, WTS sets the maximum permissible similarity gap of 0.55, whereas EC-spLLM sets the semantic matching threshold $\theta = 0.4$, similarity filtering threshold $\lambda = 0.4$, and a semantic entropy threshold $\omega = 0$. Furthermore, prompts employed in EC-spLLM are provided in the code base².

4.2 Main Results

Table 1 compares the GA usage between EC-spLLM and WTS. In WTS, the golden answer is required for each query to establish a KG [39]. As a result, WTS has a constant GA usage rate of 1.0. In contrast, EC-spLLM employs the HDEC mechanism, which selectively choose between the golden answer and the reliable LLM-generated answer for the final response and DKG evolution, significantly reducing its dependence on golden answers. As shown, EC-spLLM exhibits significantly lower GA usage across all datasets compared to WTS. Specifically, the average utilization rates for GPT-4.1 and DeepSeek-v3 under the EC-spLLM framework are 36.6% and 29.5%, respectively, representing reductions of 63.4% and 70.5%.

Table 1: GA usage comparison between EC-spLLM and WTS on different datasets.

Domain	Dataset	WTS	EC-spLLM (GPT-4.1)	EC-spLLM (DS-v3)
Biology	BioASQ	1.0	0.38 _{↓0.62}	0.28 _{↓0.72}
Sociology	TweetQA	1.0	0.323 _{↓0.677}	0.30 _{↓0.70}
Science	SimpleQA	1.0	0.446 _{↓0.554}	0.371 _{↓0.629}
Medical	PubMedQA	1.0	0.368 _{↓0.632}	0.324 _{↓0.676}
General	NQ-open	1.0	0.313 _{↓0.687}	0.21 _{↓0.79}
Average		1.0	0.366 _{↓0.634}	0.295 _{↓0.705}

However, how does EC-spLLM, which uses only a small amount of golden answers, perform in terms of accuracy? As shown in Table 2, EC-spLLM achieves the second-best results on three datasets and outperforms all baselines on the NQ-open. On the PubMedQA dataset, EC-spLLM ranks first when using GPT-4.1 and second when using DeepSeek-v3. More specifically, EC-spLLM achieves 99.74% and 94.71% of WTS’s accuracy while utilizing only 36.6% and 29.5% of the golden answers, respectively, demonstrating strong performance with less reliance on golden answers. Notably, the

¹<https://www.trychroma.com/>

²<https://github.com/Bamboowine0/EC-spLLM>

Table 2: Accuracy(%) comparison of EC-spLLM and baselines on different datasets.

Backbone	Domain	Biology	Emotional Sociology	Science	Medical	General	Average
	Dataset Method	BioASQ	TweetQA	SimpleQA	PubmedQA	NQ-open	
GPT-4.1	Query Directly	82.9	85.6	31.2	38.9	71.4	62.0
	Zero-shot-CoT	81.7	85.4	29.2	37.3	69.2	60.56
	CoT	83.2	85.0	34.1	40.6	71.8	62.94
	ToG	80.1	86.1	38.3	38.3	72.1	62.98
	EC-spLLM	86.4	86.8	52.2	54.9	76.7	71.4
	WTS	87	88.9	57.3	49.7	75	71.58
Deepseek-v3	Query Directly	79.2	82.3	20.1	33.9	64	55.9
	Zero-shot-CoT	74.2	82.2	21.5	34.9	62.4	55.04
	CoT	74.9	81.5	24.8	23.4	65.9	54.1
	ToG	74.9	82.5	24.6	34.3	65.2	56.3
	EC-spLLM	81.1	83.2	38.1	41.7	67.5	62.32
	WTS	85	84.6	47.5	44.6	67.3	65.8

results on NQ-open and PubmedQA indicate that EC-spLLM can sometimes even surpass WTS, which we attribute to EC-spLLM’s stronger retrieval capabilities enhanced by epistemic cognition. WTS attains the highest accuracy across three datasets, driven by its extensive use of golden answers and the fact that the knowledge integrated into the DKG is error-free. However, the cost of such performance is the use of a golden answer for every query, which is impractical in most real-world applications.

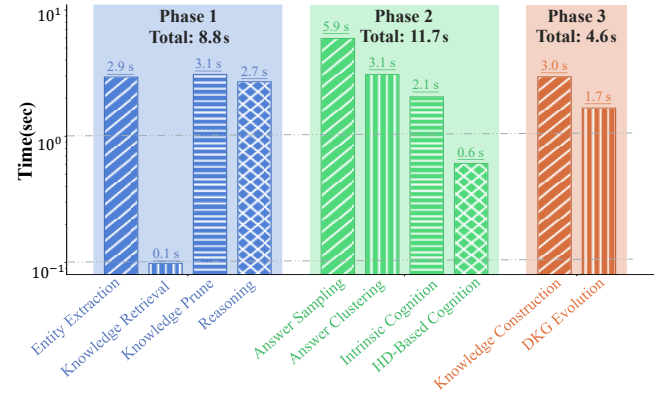
When comparing EC-spLLM to Query Directly, we observe a substantial improvement. Across the five datasets, EC-spLLM achieves an average accuracy gain of 9.4% and 6.42% over direct LLM querying, with the most significant improvements reaching 21.0% and 18.0% on the SimpleQA. This highlights the positive impact resulting from accumulated knowledge.

EC-spLLM also demonstrates significant improvements over both CoT variants. Compared to Zero-shot-CoT, EC-spLLM achieves up to 23.0% and 16.6% accuracy gains on the SimpleQA. Against standard CoT, EC-spLLM reaches maximum improvements of 18.1% on SimpleQA and 18.3% on PubmedQA. It is worth noting that EC-spLLM and CoT are not mutually exclusive. As a prompt engineering strategy, CoT can be continuously integrated into EC-spLLM to further enhance its capabilities.

Next, compared to ToG, EC-spLLM achieves maximum accuracy improvements of 16.6% on the PubmedQA and 13.5% on the SimpleQA. The performance gap can largely be attributed to ToG relying on a general-purpose knowledge graph, which often struggles to retrieve relevant knowledge for domain-specific questions, thereby limiting its effectiveness. Notably, when no knowledge is retrieved, ToG, WTS, and EC-spLLM use the same prompt as Query Directly. While this setup removes some of the advantages ToG may gain through CoT, it ensures the fairness of the experiments by maintaining consistent conditions across all methods.

In addition, we measured the time taken by ten stages of EC-spLLM (using the OpenAI API) across five datasets, with average results shown in Figure 3. Among the three phases, Phase 2 consumes the most time, mainly due to the answer sampling stage. Since 10 samples are needed, the time consumption of this stage is

primarily caused by the delay in API requests. To ameliorate this limitation, we employed multithreading, reducing the time spent on answer sampling to 5.9 seconds. This is only 3.2 seconds longer than the knowledge-augmented reasoning stage, which involves just a single API request and takes 2.7 seconds. Other stages, such as knowledge retrieval, hallucination detection-based cognition, and DKG evolution, which do not require network requests, have relatively small time costs. Notably, the knowledge retrieval stage

**Figure 3: The time consumption of EC-spLLM’s stages and phases.**

shows the most significant time advantage, owing to the use of multithreading techniques in its implementation. Since all other stages require just one API request, their time consumption is similar. Overall, EC-spLLM’s runtime is largely determined by API request time. If the time for API requests can be reduced, the overall time cost of EC-spLLM will decrease further.

4.3 Ablation Study

How does the KG influence the performance of the EC-spLLM?

Figure 4 illustrates the variation of EC-spLLM’s GA usage and accuracy improvement relative to Query Directly across five datasets

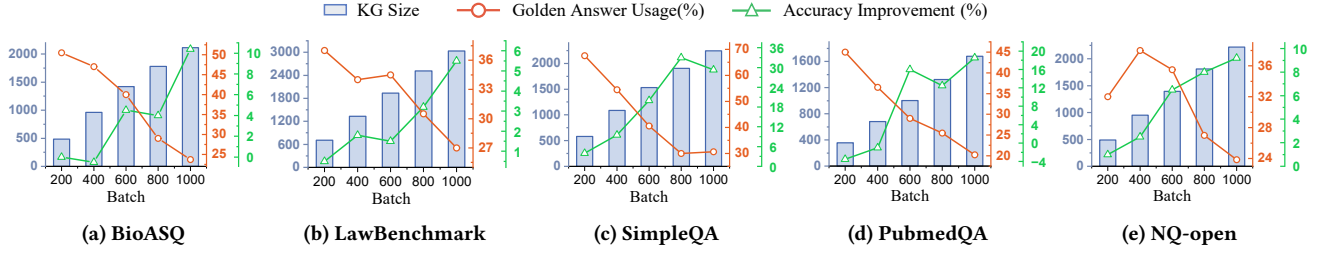


Figure 4: EC-spLLM performance changes with batch size and KG size on each dataset.

using GPT-4.1. At the outset, EC-spLLM demonstrates high GA usage but limited accuracy improvement across all datasets. During this initial stage, its accuracy improvement hovers around zero, as EC-spLLM lacks sufficient knowledge and thus performs comparably to Query Directly. With the expansion of the KG, EC-spLLM's accuracy improvement increases, while GA usage gradually declines. Although minor fluctuations are observed, the overall trend is evident. Broadly, as knowledge accumulates within the KG, EC-spLLM becomes less reliant on expert assistance and increasingly capable of answering queries autonomously. This progression suggests that, with sustained expert "education", EC-spLLM evolves toward a more self-sufficient, expert-like state, thereby validating the effectiveness of the proposed approach.

How does the knowledge extracted from LLM answers influence the performance of the EC-spLLM? We conducted experiments with EC-spLLM under a setting where LLM-generated answers were excluded as a knowledge source, and compared its performance against the version that utilizes knowledge from LLM answers. The results are shown in Figure 5.

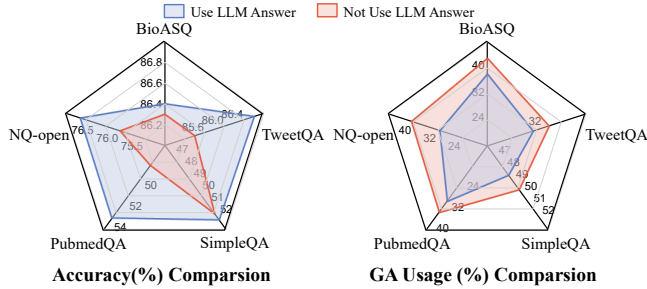


Figure 5: Comparison of accuracy(%) and golden answer usage(%) for EC-spLLM under two different knowledge construction strategies.

In this setup, EC-spLLM does not extract any knowledge from what it deems as reliable LLM answers. This means that the DKG constructed by EC-spLLM only contains knowledge derived from golden answers. However, the results demonstrate that incorporating knowledge extracted from LLM answers leads to an average improvement of 1.52% in accuracy, while reducing the usage of golden answers by an average of 4%. These findings validate the positive effect of bridging expert knowledge with the LLM's intrinsic knowledge.

5 Conclusion

In this paper, we introduce EC-spLLM, a novel evolutionary framework designed to reduce the reliance on costly expert knowledge in building KG-based specialized LLMs by instilling them with epistemic cognition. At the heart of our framework is the HDEC mechanism, which evaluates answer reliability via LLM's intrinsic cognition and hallucination detection. Guided by HDEC, EC-spLLM can selectively choose between the expert-provided golden answer and the reliable LLM-generated answer for the final response and DKG evolution. This capability drastically reduces dependence on expert knowledge while seamlessly bridging the gap between it and the LLM's internal knowledge. Rigorous evaluation results confirm our approach reduces expert reliance by over 70% while nearly matching SOTA accuracy, highlighting a promising path toward more autonomous and economical specialized AI.

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A Algorithm

In this section, we present the overall algorithmic framework of EC-spLLM. The detailed process is illustrated in Algorithm 1.

B Experiment Detail

B.1 Dataset

B.1.1 BioASQ. The BioASQ dataset [27] originates from the "Biomedical Semantic Indexing and Question Answering" challenge. It is a large-scale benchmark aimed at testing systems' capabilities in answering biomedical domain-specific questions. Questions are designed by biomedical experts and encompass four types: "yes/no," "factoid," "list," and "summary," with answers derived primarily from PubMed articles. For our experiments, we selected Task B from the

Algorithm 1: EC-spLLM workflow

Input: Question q , Domain Knowledge Graph $\mathcal{G}$, Max depth D , Entropy Threshold Ω

Output: Answer α_q

```

1  $\mathcal{K}_q \leftarrow \{\}$ ;
2  $d \leftarrow 1$ ;
3 # Phase 1: DKG-Enhanced LLM Generation
4  $\mathcal{E}_q \leftarrow \text{entities\_extraction}(q)$ ;
5  $\mathcal{E}_q^{[1]} \leftarrow \mathcal{E}_q$ ;
6 while  $d \leq D$  do
7    $\mathcal{K}_q^{[d]} \leftarrow \text{knowledge\_retrieval}(\mathcal{E}_q^{[d]}, \mathcal{G})$ ;
8    $\tilde{\mathcal{K}}_q^{[d]} \leftarrow \text{knowledge\_prune}(\mathcal{K}_q^{[d]}, q)$ ;
9    $\tilde{\mathcal{K}}_q \leftarrow \bigcup_{i=1}^d \tilde{\mathcal{K}}_q^{[i]}$ ;
10   $\alpha_q^{[d]} \leftarrow \text{LLM\_reasoning}(q, \tilde{\mathcal{K}}_q)$ ;
11  # Phase 2: Hallucination Detection-Based Epistemic Cognition
12   $\mathcal{S}_q \leftarrow \text{answer\_sampling}(q, \tilde{\mathcal{K}}_q)$ ;
13   $\mathcal{C}_q \leftarrow \text{answer\_clustering}(\mathcal{S}_q)$ ;
14   $P_{\text{Unk}} \leftarrow \text{compute\_uncertainty}(\mathcal{C}_q)$ ;
15   $\Phi_q \leftarrow \text{compute\_entropy}(\mathcal{C}_q)$ ;
16  if  $P_{\text{Unk}} > 0.5$  or  $\Phi_q > \Omega$  then
17     $\mathcal{E}_q^{[d+1]} \leftarrow \text{update\_entities}(\tilde{\mathcal{K}}_q^{[d]})$ ;
18     $d \leftarrow d + 1$ ;
19    continue;
20  else
21    break;
22 # Phase 3: LLM-Driven DKG Evolution
23 if  $d \leq H$  then
24   # use LLM answer  $\alpha_q$ 
25    $\mathcal{K}_q^+ \leftarrow \text{knowledge\_construction}(q, \alpha_q)$ ;
26    $\mathcal{G} \leftarrow \text{DKG\_evolution}(\mathcal{G}, \mathcal{K}_q^+)$ ;
27 else
28   if  $H < d \leq D$  then
29     # search laborious, use gold answer  $\alpha_G$ 
30      $\mathcal{K}_q^+ \leftarrow \text{knowledge\_construction}(q, \alpha_G)$ ;
31      $\mathcal{G} \leftarrow \text{DKG\_evolution}(\mathcal{G}, \mathcal{K}_q^+)$ ;
32   else
33     # maximum search depth exceeded, use gold answer
34      $\alpha_G$ 
35      $\mathcal{K}_q^+ \leftarrow \text{knowledge\_construction}(q, \alpha_G)$ ;
36      $\mathcal{G} \leftarrow \text{DKG\_evolution}(\mathcal{G}, \mathcal{K}_q^+)$ ;

```

2024 BioASQ challenge, which includes all question-answering tasks from BioASQ. Specifically, we focused on the factoid question-answering tasks, as they require deep and specialized knowledge in biological and medical concepts. From a total of 1,302 factoid type questions, we randomly sampled 500 questions for experimental analysis.

B.1.2 TweetQA. TweetQA [34] is a social media question-answering dataset introduced by the research team at UCSB. The dataset is

sourced from tweets posted by major news organizations on Twitter (e.g., CNN, BBC, NYT), ensuring the authenticity and reliability of the data sources. Each entry in TweetQA consists of a tweet context, a question, and a corresponding answer, where the relevant information required to answer the question can be found within the tweet itself. For our evaluation, we utilized this dataset to assess the capability of EC-spLLM specifically within the social network domain. We selected 500 questions from the validation set for analysis in our experiments.

B.1.3 SimpleQA. SimpleQA [31] is an open-source factual evaluation benchmark developed by OpenAI to assess the accuracy of large language models in answering short, fact-based questions. The benchmark includes 4,326 questions that have been independently verified by two raters, with unique answers that do not change over time. The questions span more than 10 topics, including science, history, sports, and entertainment. For our experiments, we focused on the type that has the most questions—the science category, which contains 858 questions, and randomly sampled 500 questions for testing purposes.

B.1.4 PubMedQA. PubMedQA [11] is a question-answering dataset specifically created for the medical research domain. The task is to answer a research question with "yes", "no", or "maybe" based solely on the content of a corresponding PubMed abstract. To fully display the performance of knowledge augmentation, we adopt a no-given-context and reasoning-free setting and use the labeled 500 Q&A pairs in the experiment. In addition, to prevent the model from using "maybe" as its answer when it doesn't know the answer, an "unknown" option was added to the list of possible answers. Similarly, we randomly sampled 500 questions from this dataset for our experimental evaluation.

B.1.5 NQ-open. NQ-open [14] is the open-domain question-answering component of the Natural Questions dataset, which contains real user queries sourced from the Google search engine. For each question, NQ-open provides one or more possible answers. To ensure a precise evaluation of model responses, we filtered the dataset to retain 2,076 questions with only a single correct answer. From this subset, we randomly sampled 500 questions for our experimental analysis.

B.2 Data Rephrase

In EC-spLLM, specialized question answering, such as legal consultations or medical inquiries, is assumed to be its working scenario. In such contexts, the same question may be phrased in different ways, yet the required knowledge for answering remains the same. For instance, different patients might ask a doctor, "I am allergic to penicillin; can I take amoxicillin?" and "I want to take amoxicillin to alleviate my symptoms, but I am allergic to penicillin; is it safe?" The ability of EC-spLLM can only be reasonably evaluated when tested on data exhibiting such characteristics. However, in order to improve data quality, the questions in most QA datasets are often semantically disjoint. Therefore, we perform rephrasing on randomly sampled questions from various datasets, and the overall process is as follows:

(1). For each question q , we prompt ‘chatgpt-4o-latest’ to generate a rephrased version using prompt 1 (Take the BioASQ dataset as an example):

Prompt 1: Data Rephrase

•System Prompt:

You are a professor in the field of [Biology] with extensive and accurate [Biology] knowledge. Your task is to rephrase the given questions while preserving the original meaning as much as possible and altering the expression as much as possible.

•User Prompt:

Please rewrite a short-answer question:{question}. Please note that the answer is {answer}. Make sure that after rewriting, the answer remains correct.

...In-Context Few-shot...

(2). The rephrased question $\hat{q}$ and the original question q are embedded into a high-dimensional space using the Qwen3-Embedding-0.6B model. We then calculate their cosine similarity Sim_q . If Sim_q falls within the range (0.75, 0.95), the rephrased question $\hat{q}$ is accepted; otherwise, a new rephrasing attempt is triggered for q .

(3). If the rephrasing fails to meet the similarity constraint after three attempts, we perform a final rephrasing using ‘gpt-4.5-preview’, and adopt the result as the final version.

For each dataset, we extract 500 questions and randomly mix them with their rephrased counterparts to form the final test set. Notably, all datasets used in our experiments are available in the EC-spLLM code repository.

C HDEC’s Prediction Capability

We conducted a statistical analysis comparing the judgments made by the HDEC with the correctness of LLM answers on the experiment data of GPT-4.1. Figure 6 illustrates the result. Here, we define

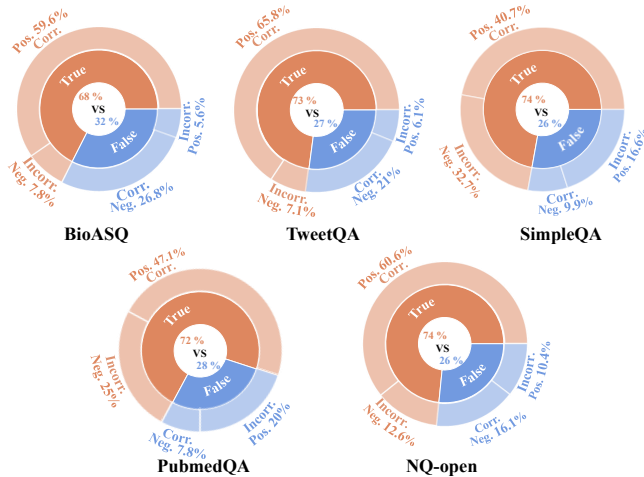


Figure 6: Alignment of HDEC predictions with LLM answer correctness.

a "true prediction" as an instance where HDEC’s assessment aligns

with the answer’s validity: positive for a correct answer, and negative for an incorrect one. Across five datasets, the HDEC achieved a peak prediction accuracy of 74% and an average of 72%.

Errors in prediction can be primarily attributed to limitations in the LLM itself, such as deficiencies in its internal reasoning ability or lack of knowledge. It is important to note that the HDEC is not designed to address shortcomings inherent in the LLM, as these issues lie within the scope of the LLM’s architecture and training. Consequently, the HDEC operates under certain performance constraints. Moreover, the hallucination problem remains a persistent challenge for LLMs, with no existing method can achieve 100% accuracy in detecting such errors. HDEC is no exception to this limitation.

D GA matching performance

Aligning EC-spLLM’s use of golden answers with LLM output correctness is crucial to the quality of the constructed KG. Figure 7 depicts their alignment on five datasets. Our primary concern is instances where the LLM’s answer is incorrect, yet no golden answer is employed to override it; we term these cases false positives. Such false positives lead our method to erroneously integrate incorrect knowledge into its KG. Across the five datasets, the average false positive rate is 10.8%, with a minimum of 5.1% on the BioASQ dataset. Importantly, the PubmedQA datasets exhibit relatively high

	BioASQ	TweetQA	SimpleQA	PubmedQA	NQ-open	
Use GA	29.7%	24.5%	11.3%	9.2%	18.0%	0.6
	8.3%	8.0%	33.3%	26.7%	13.3%	0.5
Not Use GA	56.7%	62.3%	39.3%	45.7%	58.7%	0.4
	5.1%	5.2%	16.0%	18.3%	9.7%	0.3
						0.2
						0.1

* The sum of each column is 100%

Figure 7: Alignment of golden answer usage with LLM answer correctness.

false positive rates. This is largely due to its multiple-choice format, which constrains LLM responses and lowers semantic entropy. The presence of answer options also introduces heuristic cues, prompting the LLM to "guess" the most plausible choice rather than admit incapability—contributing to an elevated false positive rate. In contrast, the other three open-domain datasets show a much lower average false positive rate of just 9%. This demonstrates the reliability of EC-spLLM’s knowledge accumulation in unconstrained response settings, which more closely align with practical application scenarios.

E Retrieval Mechanism Impact

To explore the performance impact of employing fuzzy search versus exact matching for entities—a choice mentioned in section 3.2—we conducted experiments on GPT-4.1 to measure the resulting effects on EC-spLLM.

Table 3: The Accuracy(%) and GA usage(%) of EC-spLLM with different retrieval mechanism

Dataset \ Metric	Exact Match		Fuzzy Search	
	Acc	GA usage	Acc	GA usage
BioASQ	85.8	43.8	86.4 _{↑0.6}	38.0 _{↓5.6}
TweetQA	85.0	33.7	86.8 _{↑1.8}	32.3 _{↓1.4}
SimpleQA	49.8	51.6	52.2 _{↑2.4}	44.6 _{↓7.0}
PubmedQA	48.6	43.6	54.9 _{↑6.3}	36.8 _{↓6.8}
NQ-open	72.8	44.5	76.7 _{↑3.9}	31.3 _{↓13.2}

As shown in Table 3, EC-spLLM using fuzzy search outperforms across all five datasets. Compared to using exact match, fuzzy search results in an average improvement of 3% in accuracy and a reduction of 6.8% in GA usage. This indicates that using fuzzy search as the retrieval strategy for EC-spLLM is more effective.

F Prompt

EC-spLLM relies on the LLM to complete six distinct stages. Below, we illustrate the prompt template employed by EC-spLLM with a specific example used on the BioASQ dataset. More detail can be found in the code base.

Prompt 2: Entities Extraction

- System Prompt: You are a [Biological] assistant to carry out [Biological] Name Entity Recognition from a question and output JSON like {entities: [entity1, entity2,...]}. Since the entity you extract is used to retrieve relevant information, you will generally pay more attention to rarer entities.
- User Prompt: Get the meaningful entities in the question: {Question}. The answer should be like this:
...In-Context Few-shot...

Prompt 3: Knowledge Prune

- System Prompt: You are an excellent [biology] teaching assistant with extensive and accurate [biological] knowledge. You are asked to evaluate some triplets with validity scores to help your professor answer questions. The range of validity scores is between 1.0 and 0.0. If you think a triple is more helpful in answering this question, the higher it scores, and vice versa. A score of 1.0 means that the triple can answer the question directly, and as the score decreases (0.95, 0.9, 0.8...) the triple helps to answer questions less. You are asked to output your results in JSON format.

- User Prompt: Based on the question: [Question], score the triples: [Triples]. The score is between 0 and 1 and the order is from high to low. Here is an example:
...In-Context Few-shot...

Prompt 4: Knowledge-Augmented Reasoning

- System Prompt: You are a [biologist]. Please provide concise and accurate answers to the following questions. The output must be a dict in JSON format with only two keys: answer and support_info and your output must be like {answer:xxx,support_info:xxx}.
- User Prompt: Based on your own knowledge and the <knowledge triple>, please answer the question concisely. Q: [{Question}], knowledge triple: [{Triple}], A:?
Example:
...In-Context Few-shot...

Prompt 5: Answer Clustering

- System Prompt: You are a logician. Here, we define a relation called "entail". Suppose there are two sentences A and B. If A can logically imply B (proving B to be true), then we say A entails B. If A and B mutually entail each other, then A and B can be considered semantically identical and classified as the same category. For example, the sentences "The capital of France is Paris" and "Paris is the capital of France" mutually entail each other, meaning these two sentences are equivalent and can be considered as belonging to the same category.
- User Prompt: Now, I will provide you with some sentences, they are the answers for the question: {Question}. Please cluster the sentences provided below and return the results in JSON format (note: please return directly in JSON format without outputting any other content!);{Sentences}; Below is a reference example of the return result (only for style reference, the content is unrelated to your task):
...In-Context Few-shot...

Finally, prompt 6 is presented, which is used to evaluate the correctness of answers to open-domain questions.

Prompt 6: Answer Evaluation

- System Prompt: Null
- User Prompt: I will provide you with a question, a answer generated by llm and a real answer. Please judge whether the llm answer is correct. You just need to return a JSON with one key: 'result', and the value is a boolean value. If the llm answer is correct, the result should be true, otherwise it should be false.
QUESTION:[{q}]
LLM-ANSWER:[{llm-answer}]
REAL-ANSWER:[{real-answer}]